

Trylen Stephens

Control Systems & Instrumentation Technology

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SUMMARY

Senior Control Systems student with three years of AI-enabled R & D experience in two different industry partnered projects. These projects include evaluating local RAG chatbots for technical education (Betaflix), and production analytics / optimization efforts for a Louisiana sawmill(Hunt/ Lasalle). My strengths include automated evaluation design, PLC programming, plant data analysis, and agentic-AI tooling.

EDUCATION

B.S Control Systems & Instrumentation Technology

Expected Spring 2027

Louisiana Tech University - Ruston, LA

- Coursework - PID tuning, Do-More PLC Programming, Measurements & Instrumentation, Applied Manufacturing Systems, AI & Machine learning
- First place in class competition for a restricted control algorithm that stabilized a thermal system
- Selected to present AI based control system to faculty

EXPERIENCE

Researcher - RAG Backed LLM Education Chatbot (Betaflix/La Tech)

Jan 2024 - Jan 2026

Louisiana Tech University - Ruston, LA

Local RAG assistant for technical training content (welding safety, aircraft maintenance)

- Developed, tested, and scored local RAG pipelines (LlamaIndex, ChromaDB, Ollama) across embedding and local model configurations on hand crafted benchmarks.
- Assisted in development of evaluation environments, including automated grading and score sheet verification
- Built early local LLM grading tools with NLI/entailment methods
- Responsible for technical documentation and review before reporting results to the company

Researcher - Sawmill Operation Analytics (Hunt-Lasalle/ La Tech)

Jan 2026 - Present

Louisiana Tech University - Ruston, LA

Data and process analytics for downtime, trim loss, and plant operations optimization

- Built and integrated a process model of the mill for live signal visualization tool on production dashboard
- Developed agentic trimloss analysis environment and managed updates and releases on production dashboard
- Created cost reports, yearly signal charts, machine center failure ranking, rule-search pipelines

TECHNICAL SKILLS

PLC/HMI - Do-More PLC, C-More HMI, ladder logic

Controls - PID Tuning, broad sensor integration, state machine logic implementation

Programming- Python, Github, Arduino/C++

AI/Evaluation - RAG, local LLMs, Cursor, Claude Code & other agentic harnesses